THEORY REPORT

Assignment 5

Title: Analyze the Permissions Requested by an Android Application and Identify Excessive or Risky Permissions

1. Aim

To analyze the permissions requested by an Android application and identify permissions that may be excessive, unnecessary, or security-sensitive.

2. Objectives

1. To understand the Android permission model.
2. To identify permissions requested by an application.
3. To classify permissions according to their security and privacy impact.
4. To detect unnecessary or excessive permissions.
5. To understand the risks associated with sensitive permissions.

3. Introduction

Android applications require permissions when they need access to protected device resources or user data. Permissions help protect sensitive information such as location, contacts, camera, microphone, storage, and device-related information.

An application should request only the permissions required for its intended functionality. Excessive permissions increase the application's **attack surface** and may create privacy and security risks. Therefore, analyzing an application's permissions is an important part of Android security assessment.

4. Android Permission Model

Android permissions can broadly be understood as:

- **Normal permissions:** Permissions associated with relatively low-risk operations and generally handled automatically by the system.
- **Dangerous permissions:** Permissions that provide access to sensitive user data or device capabilities and generally require runtime user approval.
- **Signature-level permissions:** Permissions intended for applications signed with the appropriate certificate or otherwise restricted by the system.

The exact behavior of permissions depends on the Android version and the particular permission.

5. Permission Analysis

Permissions can be examined from the application's manifest or using Android debugging tools.

For an authorized test device, ADB can be used to inspect an installed package:

adb shell dumpsys package <package-name>

The output can contain information about permissions associated with the application.

A permission should be considered potentially excessive when the application's functionality does not clearly require it.

Examples

Permission / Capability	**Risk Level**	**Possible Risk**
Camera	Medium/High	Unauthorized image capture
Microphone	High	Unauthorized audio recording
Location	High	Location tracking and privacy loss
Contacts	High	Exposure of personal contacts
SMS	High	Access to sensitive messages or OTP-related data
Phone-related access	High	Exposure of communication information
Notifications	Medium	Access to notification content depending on mechanism
Internet	Generally lower	Enables network communication; risk depends on application behavior

6. Identifying Excessive Permissions

The following questions can be used during analysis:

1. Is the permission necessary for the application's primary functionality?
2. Does the application request access to sensitive personal information?
3. Is the permission required continuously or only for a particular feature?
4. Could the same functionality be implemented with a less-privileged API?
5. Does the application's stated purpose justify the requested access?

For example, a calculator application requesting microphone, contacts, SMS, and location access would require further investigation because these permissions are not normally necessary for basic calculator functionality.

7. Security Risks

Excessive permissions may lead to:

- Privacy violations
- Unauthorized access to sensitive information

- Data leakage
- Increased attack surface
- Misuse of device capabilities
- Greater impact if the application is compromised

Applications should follow the **Principle of Least Privilege**, requesting only the minimum permissions necessary.

8. Security Recommendations

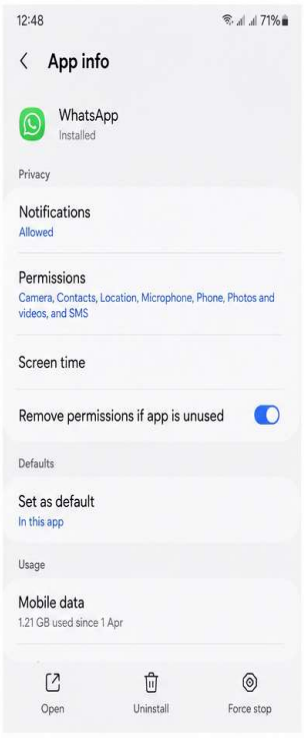
- Review permissions before installing applications.
- Remove unnecessary permissions where Android allows the user to do so.
- Request sensitive permissions only when required.
- Explain why sensitive permissions are needed.
- Avoid requesting broad access when a more limited API is available.
- Regularly review permissions granted to installed applications.
- Developers should remove unused permissions from the application manifest.

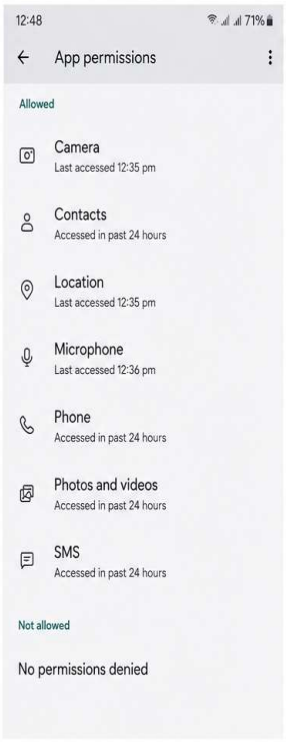
9. Result

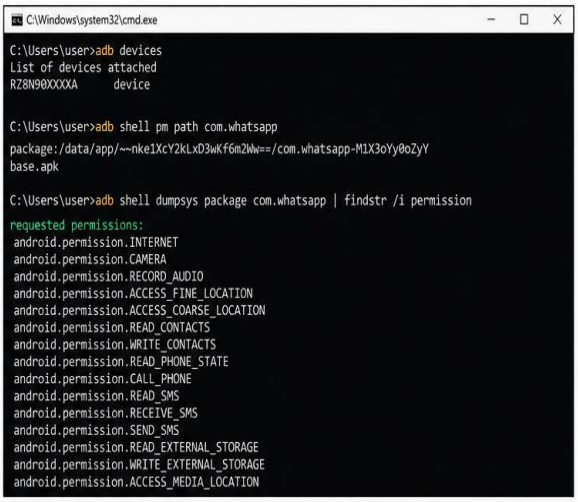
The permissions requested by an Android application were analyzed and classified according to their potential security and privacy impact. Permissions that were unnecessary for the application's intended functionality were identified as potentially excessive or risky.

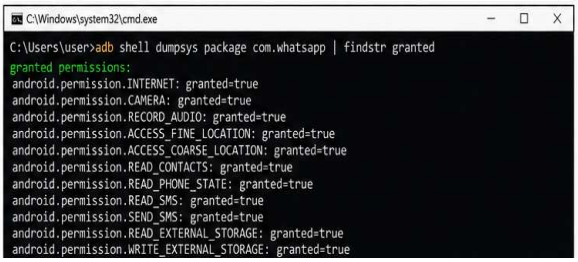
10. Conclusion

Permission analysis is an important part of Android application security. Sensitive permissions can provide access to valuable user information and device capabilities. Therefore, applications should follow the **least-privilege principle** and request only permissions that are essential for their functionality. Regular permission analysis helps reduce privacy risks, limit the attack surface, and improve the overall security of Android applications.


12:48
71%
App info
WhatsApp
Installed
Privacy
Notifications
Allowed
Permissions
Camera, Contacts, Location, Microphone, Phone, Photos and videos, and SMS
Screen time
Remove permissions if app is unused
Defaults
Set as default
In this app
Usage
Mobile data
1.21 GB used since 1 Apr
Open
Uninstall
Force stop


12:48
71%
App permissions
Allowed
Camera
Last accessed 12:35 pm
Contacts
Accessed in past 24 hours
Location
Last accessed 12:35 pm
Microphone
Last accessed 12:36 pm
Phone
Accessed in past 24 hours
Photos and videos
Accessed in past 24 hours
SMS
Accessed in past 24 hours
Not allowed
No permissions denied


C:\Windows\system32\cmd.exe
C:\Users\user>adb devices
List of devices attached
RZ8N90XXXXA device
C:\Users\user>adb shell pm path com.whatsapp
package:/data/app/~~nke1XcY2kLxD3wKf6m2Ww==/com.whatsapp-M1X3oYy0oZyY
base.apk
C:\Users\user>adb shell dumpsys package com.whatsapp | findstr /i permission
requested permissions:
android.permission.INTERNET
android.permission.CAMERA
android.permission.RECORD_AUDIO
android.permission.ACCESS_FINE_LOCATION
android.permission.ACCESS_COARSE_LOCATION
android.permission.READ_CONTACTS
android.permission.WRITE_CONTACTS
android.permission.READ_PHONE_STATE
android.permission.CALL_PHONE
android.permission.READ_SMS
android.permission.RECEIVE_SMS
android.permission.SEND_SMS
android.permission.READ_EXTERNAL_STORAGE
android.permission.WRITE_EXTERNAL_STORAGE
android.permission.ACCESS_MEDIA_LOCATION


C:\Windows\system32\cmd.exe
C:\Users\user>adb shell dumpsys package com.whatsapp | findstr granted
granted permissions:
android.permission.INTERNET: granted=true
android.permission.CAMERA: granted=true
android.permission.RECORD_AUDIO: granted=true
android.permission.ACCESS_FINE_LOCATION: granted=true
android.permission.ACCESS_COARSE_LOCATION: granted=true
android.permission.READ_CONTACTS: granted=true
android.permission.READ_PHONE_STATE: granted=true
android.permission.READ_SMS: granted=true
android.permission.SEND_SMS: granted=true
android.permission.READ_EXTERNAL_STORAGE: granted=true
android.permission.WRITE_EXTERNAL_STORAGE: granted=true